

Functional Requirement Document for BMI Calculator

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1. Introduction:

The functional requirement document provides a complete description of the BMI Calculator web application. It outlines the structure, behavior, and user interaction across the homepage, calculator page, and informational content. This document serves as a guide for developers, testers, and stakeholders to understand the expected functionality and flow of the application.

1.1 Purpose:

The purpose of this document is to define the functional requirements for the BMI Calculator website. The application allows users to calculate their Body Mass Index (BMI) based on height and weight inputs, view categorized health feedback, and understand the importance of BMI. This document ensures clarity in development and testing by describing how users interact with the system and how the system responds.

1.2 Intended Audience:

- Development Team
- QA Team
- Business Analyst
- UI/UX Design Team

2. Functional Requirements:

- When the user types the URL <https://bodymassindexcalculator1.netlify.app/> it should navigate to the Home page of BMI Calculator website.

2.1 Home Page:

- The Home Page serves as the entry point to the BMI Calculator application.
- It contains the following elements in the top section:
 - Navigation Bar with links to:
 - Home (index.html)
 - Calculator (calculator.html)
 - About (about.html)
 - A Start Button that redirects users to the BMI Calculator page.
- The page features a full-screen background image with centered alignment, creating a visually engaging experience.

- The page welcomes users with a heading: “Welcome to the BMI Calculator”
(Home page screen is shown in the **Figure 1.1**)

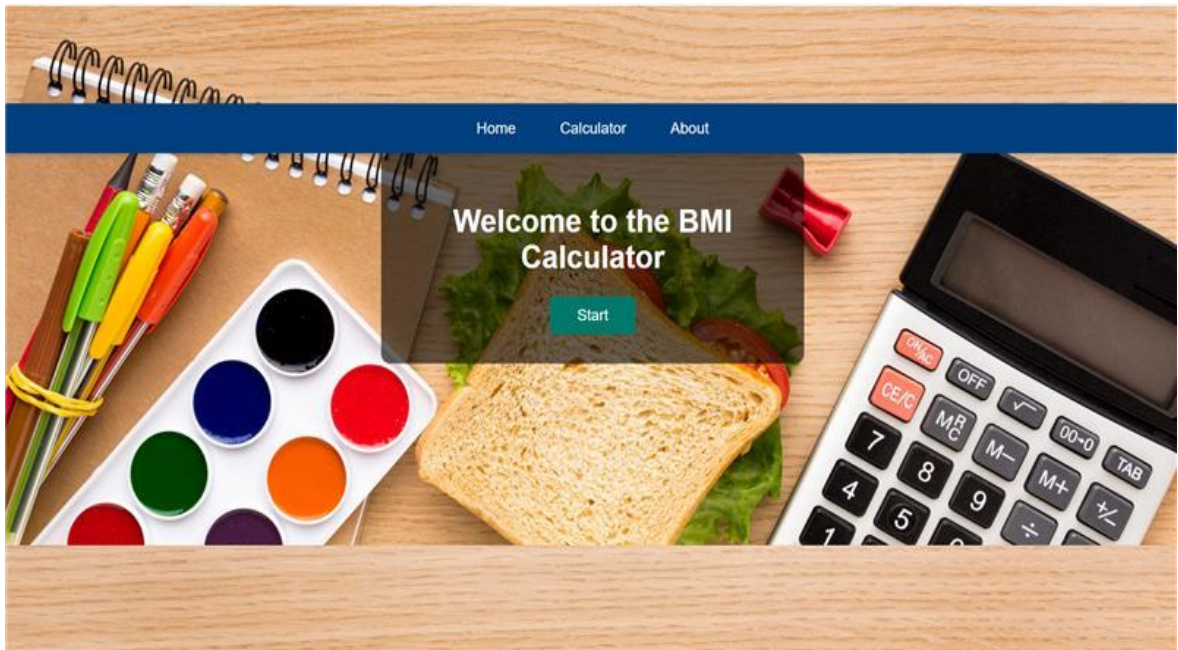


Figure 1.1

2.2 BMI Calculator Page

- The Calculator Page allows users to input their height and weight to calculate BMI.
- The page features a full-screen background image with centered alignment, creating a visually engaging experience.
(Calculator page screen is shown in the **Figure 1.2**)
- It contains the following input fields:
 - **Height Input:**
 - Textbox to enter numeric value
 - Dropdown to select unit: Centimeters or Inches
 - **Weight Input:**
 - Textbox to enter numeric value
 - Dropdown to select unit: Kilograms or Pounds
- Two buttons are provided:
 - **Calculate BMI:** Triggers BMI calculation and displays result
 - **Reset:** Clears all input fields and result display



Figure 1.2

2.2.1 BMI Calculation Logic

- On clicking Calculate BMI, the system:
 - Validates inputs for non-empty, numeric, and realistic values
 - Converts units to metric system:
 - Inches → Meters
 - Pounds → Kilograms
 - Applies BMI formula: $BMI = \text{weight (kg)} / \text{height (m)}^2$
 - Rounds the result to two decimal places

2.2.2 Result Display

- After calculation, the result is shown in the Result Box:
 - **BMI Value:** Displayed in bold
 - **Health Message:** Based on BMI category
 - **Underweight:** BMI < 18.5
 - **Normal:** BMI 18.5–24.9
 - **Overweight:** BMI ≥ 25
 - Each category includes:
 - Health advice
 - Suggested exercises
 - Recommended diet
- The result box is color-coded:
 - Yellow for Underweight

- Green for Normal
- Red for Overweight

(Result display screen is shown in the **Figure 1.3**)

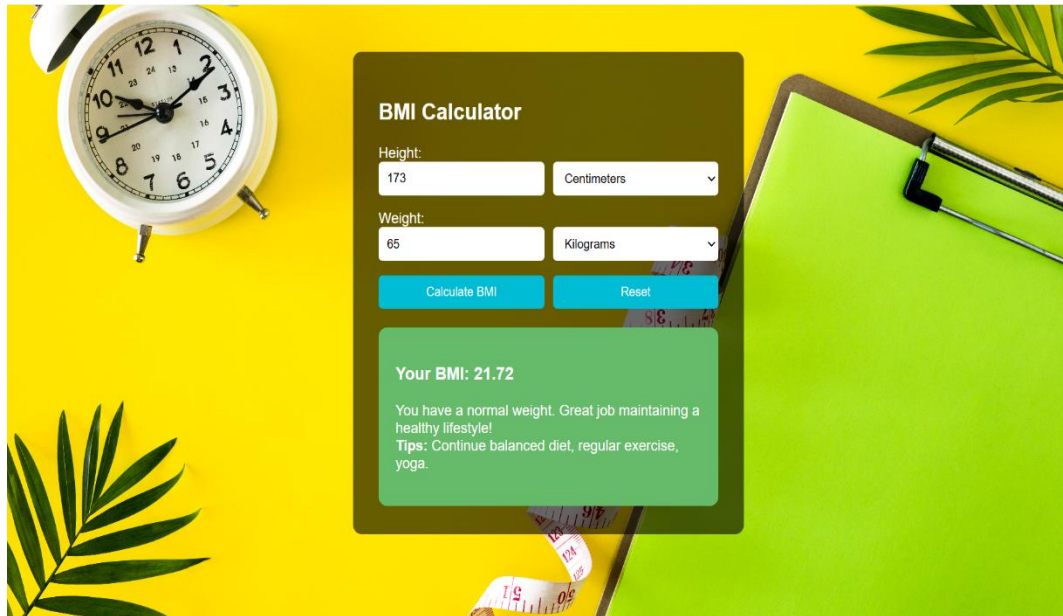


Figure 1.3

2.2.3 Input Validation

- If height or weight is missing, non-numeric, or exceeds human limits:
 - Alert messages are shown:
 - “Please enter valid height and weight values.”
 - “Height exceeds the maximum recorded human height.”
 - “Weight exceeds the maximum recorded human weight.”
 - “Both height and weight exceed the maximum recorded human limits.”

(Result display screen is shown in the **Figure 1.4**)

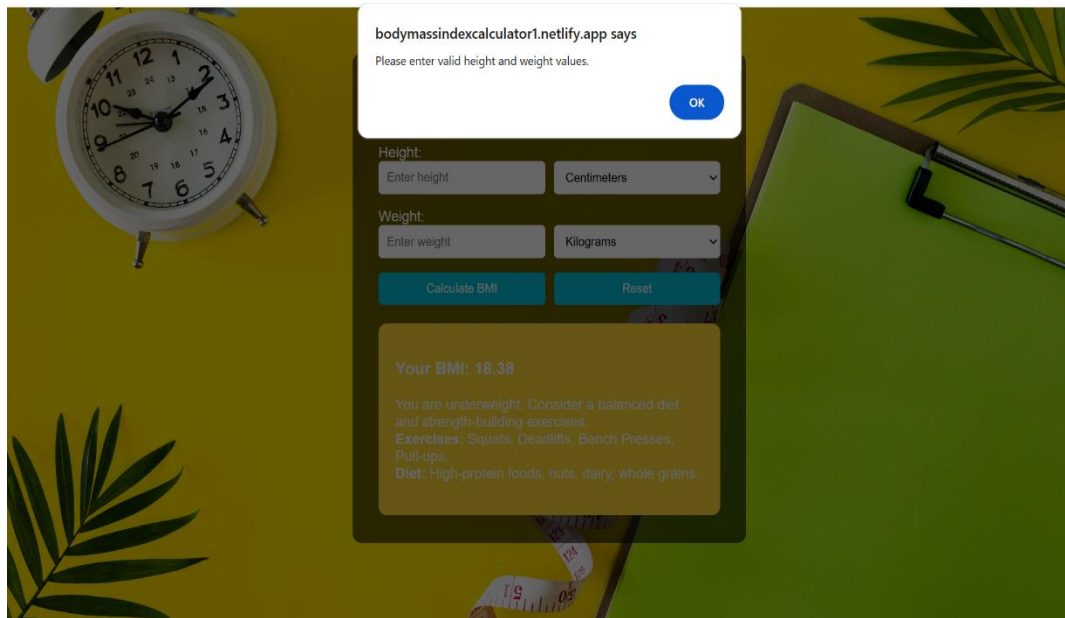


Figure 1.4

2.3 About Page

- The About Page provides educational content about BMI:
 - What is BMI?
 - Why is BMI important?
- Content is styled with a dark overlay and centered layout
- The page features a full-screen background image with centered alignment, creating a visually engaging experience.

(About page screen is shown in the **Figure 1.5**)

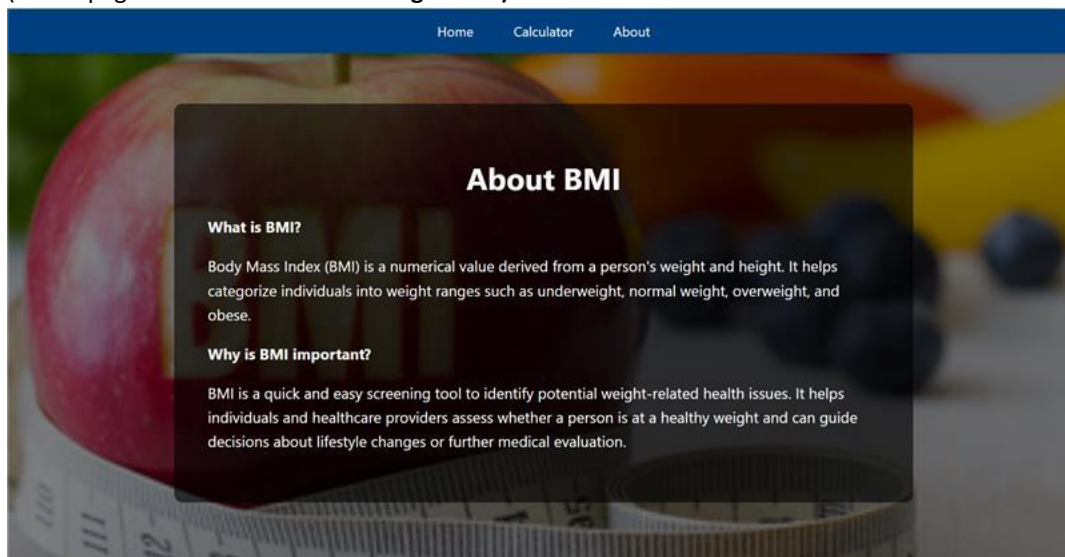


Figure 1.5